

Cryptography Exercices

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Week 1

Exercise 1.4

Let $a, b, c \in \mathbb{Z}$. Use the definition of divisibility to directly prove the following properties of divisibility (in each case, when the division is well-defined, i.e., the divisor is non-zero).

1. If $a|b$ and $b|c$, then $a|c$.
2. If $a|b$ and $b|a$, then $a = \pm b$.
3. If $a|b$ and $a|c$, then $a|(b + c)$ and $a|(b - c)$.

1. Transitivity of Divisibility

Proof. Assume $a|b$ and $b|c$. By the definition of divisibility, since $a|b$ and $a \neq 0$, there exists an integer $c_1 \in \mathbb{Z}$ such that $b = a \cdot c_1$ [Def: Divisibility]. Similarly, since $b|c$ and $b \neq 0$, there exists an integer $c_2 \in \mathbb{Z}$ such that $c = b \cdot c_2$ [Def: Divisibility].

Substitute the first equation into the second:

$$c = (a \cdot c_1) \cdot c_2 = a \cdot (c_1 c_2)$$

Since $c_1, c_2 \in \mathbb{Z}$, their product $k = c_1 c_2$ is also an integer [Prop: Closure of $\mathbb{Z}$ under multiplication]. Thus, we have $c = a \cdot k$, which, by the definition of divisibility, means $a|c$ [Def: Divisibility]. $\square$

2. Equality from Mutual Divisibility

Proof. Assume $a|b$ and $b|a$, with $a \neq 0$ and $b \neq 0$. By the definition of divisibility, we have:

$$\begin{aligned} b &= a \cdot c_1 \quad \text{for some } c_1 \in \mathbb{Z} \\ a &= b \cdot c_2 \quad \text{for some } c_2 \in \mathbb{Z} \end{aligned}$$

Substitute the expression for b from the first equation into the second:

$$a = (a \cdot c_1) \cdot c_2 = a \cdot (c_1 c_2)$$

Since $a \neq 0$, we can divide both sides by a [Prop: Division property of equality, since $a \neq 0$]:

$$1 = c_1 c_2$$

Since c_1 and c_2 are integers, the only possible pairs for (c_1, c_2) are $(1, 1)$ or $(-1, -1)$ [Prop: Divisors of 1 in $\mathbb{Z}$].

- Case 1: If $c_1 = 1$, substituting into $b = a \cdot c_1$ gives $b = a \cdot 1 = a$.
- Case 2: If $c_1 = -1$, substituting into $b = a \cdot c_1$ gives $b = a \cdot (-1) = -a$.

Therefore, we conclude that $a = \pm b$. $\square$

3. Linearity Property of Divisibility

Proof. Assume $a|b$ and $a|c$, with $a \neq 0$. By the definition of divisibility, we have:

$$\begin{aligned} b &= a \cdot k_1 \quad \text{for some } k_1 \in \mathbb{Z} \\ c &= a \cdot k_2 \quad \text{for some } k_2 \in \mathbb{Z} \end{aligned}$$

Demonstration for the Sum ($a|(b+c)$): Summing the two equations:

$$b+c = a \cdot k_1 + a \cdot k_2 = a \cdot (k_1 + k_2) \quad [\text{Factoring}]$$

Let $k_s = k_1 + k_2$. Since $k_1, k_2 \in \mathbb{Z}$, $k_s \in \mathbb{Z}$ [Prop: Closure of $\mathbb{Z}$ under addition]. Thus, $b+c = a \cdot k_s$, which means $a|(b+c)$ [Def: Divisibility].

Demonstration for the Difference ($a|(b-c)$): Subtracting the second equation from the first:

$$b-c = a \cdot k_1 - a \cdot k_2 = a \cdot (k_1 - k_2) \quad [\text{Factoring}]$$

Let $k_d = k_1 - k_2$. Since $k_1, k_2 \in \mathbb{Z}$, $k_d \in \mathbb{Z}$ [Prop: Closure of $\mathbb{Z}$ under subtraction]. Thus, $b-c = a \cdot k_d$, which means $a|(b-c)$ [Def: Divisibility]. $\square$

Exercise 1.5

Use a calculator and the method described in Lesson 2 to compute the following quotients and remainders:

1. 34787 divided by 353.
2. 238792 divided by 7843.
3. 9829387493 divided by 873485.
4. 1498387487 divided by 76348.

1. 34787 divided by 353

$$q = \lfloor \frac{34787}{353} \rfloor = \lfloor 98.5467... \rfloor = 98 \quad [\text{Def: Quotient, } q = \lfloor a/b \rfloor]$$

$$r = a - bq = 34787 - (353 \cdot 98) = 34787 - 34594 = 193 \quad [\text{Def: Remainder, } r = a - bq]$$

Result: $34787 = 353 \cdot 98 + 193$. The quotient is 98 and the remainder is 193.

2. 238792 divided by 7843

$$q = \lfloor \frac{238792}{7843} \rfloor = \lfloor 30.4449... \rfloor = 30$$

$$r = 238792 - (7843 \cdot 30) = 238792 - 235290 = 3502$$

Result: $238792 = 7843 \cdot 30 + 3502$. The quotient is 30 and the remainder is 3502.

3. 9829387493 divided by 873485

$$q = \lfloor \frac{9829387493}{873485} \rfloor = \lfloor 11253.1167... \rfloor = 11253$$

$$r = 9829387493 - (873485 \cdot 11253) = 9829387493 - 9828859005 = 528488$$

Result: $9829387493 = 873485 \cdot 11253 + 528488$. The quotient is 11253 and the remainder is 528488.

4. 1498387487 divided by 76348

$$q = \lfloor \frac{1498387487}{76348} \rfloor = \lfloor 19626.5458... \rfloor = 19626$$

$$r = 1498387487 - (76348 \cdot 19626) = 1498387487 - 1498357048 = 30439$$

Result: $1498387487 = 76348 \cdot 19626 + 30439$. The quotient is 19626 and the remainder is 30439.

Exercise 1.6

Use the Euclidean algorithm to compute the following greatest common divisors:

1. $\gcd(291, 252)$.
 2. $\gcd(16261, 85652)$.
 3. $\gcd(139024789, 93278890)$.
 4. $\gcd(16534528044, 8332745927)$.
1. $\gcd(291, 252)$ We apply the Euclidean Algorithm [Thm 3.1: Euclidean Algorithm]:

$$291 = 252 \cdot 1 + 39$$

$$252 = 39 \cdot 6 + 18$$

$$39 = 18 \cdot 2 + 3$$

$$18 = 3 \cdot 6 + 0$$

The last non-zero remainder is 3. **Result:** $\gcd(291, 252) = 3$.

2. $\gcd(16261, 85652)$ We apply the Euclidean Algorithm [Thm 3.1: Euclidean Algorithm], setting $r_0 = 85652$ and $r_1 = 16261$.

$$85652 = 16261 \cdot 5 + 4347$$

$$16261 = 4347 \cdot 3 + 3220$$

$$4347 = 3220 \cdot 1 + 1127$$

$$3220 = 1127 \cdot 2 + 966$$

$$1127 = 966 \cdot 1 + 161$$

$$966 = 161 \cdot 6 + 0$$

The last non-zero remainder is 161. **Result:** $\gcd(16261, 85652) = 161$.

3. $\gcd(139024789, 93278890)$

$$139024789 = 93278890 \cdot 1 + 45745899$$

$$93278890 = 45745899 \cdot 2 + 1787092$$

$$45745899 = 1787092 \cdot 25 + 1098599$$

$$1787092 = 1098599 \cdot 1 + 688493$$

$$1098599 = 688493 \cdot 1 + 410106$$

$$688493 = 410106 \cdot 1 + 278387$$

$$410106 = 278387 \cdot 1 + 131719$$

$$278387 = 131719 \cdot 2 + 14949$$

$$131719 = 14949 \cdot 8 + 11995$$

$$14949 = 11995 \cdot 1 + 2954$$

$$11995 = 2954 \cdot 4 + 179$$

$$2954 = 179 \cdot 16 + 70$$

$$179 = 70 \cdot 2 + 39$$

$$70 = 39 \cdot 1 + 31$$

$$39 = 31 \cdot 1 + 8$$

$$31 = 8 \cdot 3 + 7$$

$$8 = 7 \cdot 1 + 1$$

$$7 = 1 \cdot 7 + 0$$

The last non-zero remainder is 1. **Result:** $\gcd(139024789, 93278890) = 1$.

4. $\gcd(16534528044, 8332745927)$

$$16534528044 = 8332745927 \cdot 1 + 8201782117$$

$$8332745927 = 8201782117 \cdot 1 + 130963810$$

$$8201782117 = 130963810 \cdot 62 + 8459497$$

$$130963810 = 8459497 \cdot 15 + 4679365$$

$$8459497 = 4679365 \cdot 1 + 3780132$$

$$4679365 = 3780132 \cdot 1 + 899233$$

$$3780132 = 899233 \cdot 4 + 20320$$

$$899233 = 20320 \cdot 44 + 8153$$

$$20320 = 8153 \cdot 2 + 4014$$

$$8153 = 4014 \cdot 2 + 125$$

$$4014 = 125 \cdot 32 + 14$$

$$125 = 14 \cdot 8 + 13$$

$$14 = 13 \cdot 1 + 1$$

$$13 = 1 \cdot 13 + 0$$

The last non-zero remainder is 1. **Result:** $\gcd(16534528044, 8332745927) = 1$.

Exercise 1.7

Find the smallest natural B such that $345 < 2^B$. Predict the number of bits of 345 in the binary representation and find it.

Solution. The number of bits B required to represent a natural number m in binary is given by:

$$B = \lfloor \log_2 m \rfloor + 1 \quad [\text{Cor 1.1: Range of a B-bit Number, or upper bound } m < 2^B]$$

For $m = 345$:

$$B = \lfloor \log_2 345 \rfloor + 1$$

Since $2^8 = 256$ and $2^9 = 512$, we have $256 \leq 345 < 512$.

$$\log_2 256 \leq \log_2 345 < \log_2 512$$

$$8 \leq \log_2 345 < 9$$

Therefore, $\lfloor \log_2 345 \rfloor = 8$.

$$B = 8 + 1 = 9$$

The smallest natural number B such that $345 < 2^B$ is $B = 9$. This is the number of bits required.

To find the binary representation, we use the method of repeated division by 2 [Prop 1.5: Binary Representation from Repeated Division]:

$$\begin{aligned} 345 &= 2 \cdot 172 + 1 & (m_0) \\ 172 &= 2 \cdot 86 + 0 & (m_1) \\ 86 &= 2 \cdot 43 + 0 & (m_2) \\ 43 &= 2 \cdot 21 + 1 & (m_3) \\ 21 &= 2 \cdot 10 + 1 & (m_4) \\ 10 &= 2 \cdot 5 + 0 & (m_5) \\ 5 &= 2 \cdot 2 + 1 & (m_6) \\ 2 &= 2 \cdot 1 + 0 & (m_7) \\ 1 &= 2 \cdot 0 + 1 & (m_8) \end{aligned}$$

The binary representation is formed by the sequence of remainders, read from the last (m_{B-1}) to the first (m_0) :

$$345_{10} = (101011001)_2$$

Result: The smallest B is 9. The number of bits is 9. The binary representation is $(101011001)_2$.

Exercise 1.8

Let $a \in \mathbb{N}$ and $a = 9q + 13$. What are the quotient and the remainder of the division of a by 9?

Solution. The given expression for a is:

$$a = 9q + 13$$

This is close to the Euclidean Division form $a = bq' + r$ with divisor $b = 9$, but the condition on the remainder $0 \leq r < 9$ is not satisfied since $13 \not< 9$.

We must rewrite the remainder $r_{old} = 13$ by dividing it by the divisor $b = 9$ [Prop 1.4: Euclidean Division]:

$$13 = 9 \cdot 1 + 4$$

Substitute this back into the expression for a :

$$a = 9q + (9 \cdot 1 + 4)$$

$$a = 9q + 9 \cdot 1 + 4$$

$$a = 9(q + 1) + 4 \quad [\text{Factoring out } 9]$$

Let the new quotient be $q' = q + 1$ and the new remainder be $r' = 4$. The Euclidean Division is unique and requires $0 \leq r' < 9$. Since $0 \leq 4 < 9$, this condition is satisfied [Thm: Uniqueness of Quotient and Remainder]. **Result:** The quotient is $q' = q + 1$ and the remainder is $r' = 4$.

Exercise 1.9

Let $m \in \mathbb{N}$ and $m < 2^4$. Show why and how m can be represented in binary digits and how many?

Solution. Since $m \in \mathbb{N}$ and $m < 2^4$, the number m must be represented using at most $B = 4$ bits. The number of bits B is uniquely determined by the smallest integer satisfying $m < 2^B$.

1. **Number of Bits:** The number of bits B is $\lfloor \log_2 m \rfloor + 1$. Since the greatest possible value for m is $2^4 - 1 = 15$, and $\lfloor \log_2 15 \rfloor + 1 = 3 + 1 = 4$, the maximum number of bits required is 4. In general, any natural number m satisfying $m < 2^B$ can be represented with B bits (where leading zeros are allowed) [Rem: Upper Bound of a B-bit Number, $m < 2^B$].
2. **How it is Represented:** Every natural number m can be uniquely represented in base 2 (binary) as a sum of powers of 2, where the coefficients are the bits $m_i \in \{0, 1\}$.

$$m = m_0 \cdot 2^0 + m_1 \cdot 2^1 + m_2 \cdot 2^2 + m_3 \cdot 2^3 + \dots$$

For $m < 2^4$, the representation is:

$$m = m_0 + 2m_1 + 4m_2 + 8m_3$$

The bits $(m_3 m_2 m_1 m_0)_2$ are found using the ****Repeated Division by 2 Algorithm**** [Prop 1.5: Binary Representation from Repeated Division]:

$$m = 2q_0 + m_0 \quad (m_0 \text{ is the least significant bit})$$

$$q_0 = 2q_1 + m_1$$

$$q_1 = 2q_2 + m_2$$

$$q_2 = 2q_3 + m_3 \quad (q_3 \text{ must be 0 for } m < 2^4)$$

The process is guaranteed to terminate because the sequence of quotients q_i is strictly decreasing and non-negative, and for $B = 4$, the final quotient q_3 must be 0 [Thm: Termination of Repeated Division].

Week 2

Exercise 2.1

Use the Extended Euclidean Algorithm (EEA) to find integers u and v such that $au + bv = \gcd(a, b)$.

1. $\gcd(291, 252)$.

1. $\gcd(291, 252)$

Step 1: Apply the Euclidean Algorithm to find $d = \gcd(291, 252)$

$$291 = 252 \cdot 1 + 39$$

$$252 = 39 \cdot 6 + 18$$

$$39 = 18 \cdot 2 + 3$$

$$18 = 3 \cdot 6 + 0$$

The greatest common divisor is $d = 3$, the last non-zero remainder.

Step 2: Apply the Extended Euclidean Algorithm (Backward Substitution) We express the remainder 3 as a linear combination of 291 and 252 by working backward through the division steps:

$$3 = 39 - 18 \cdot 2 \quad [\text{From the third equation, isolating the remainder 3}]$$

$$3 = 39 - (252 - 39 \cdot 6) \cdot 2 \quad [\text{Substitute for 18 using the second equation}]$$

$$3 = 39 - 252 \cdot 2 + 39 \cdot 12 \quad [\text{Distribute the multiplication by 2}]$$

$$3 = 39 \cdot (1 + 12) - 252 \cdot 2 \quad [\text{Collect terms involving 39}]$$

$$3 = 39 \cdot 13 - 252 \cdot 2$$

$$3 = (291 - 252 \cdot 1) \cdot 13 - 252 \cdot 2 \quad [\text{Substitute for 39 using the first equation}]$$

$$3 = 291 \cdot 13 - 252 \cdot 13 - 252 \cdot 2 \quad [\text{Distribute the multiplication by 13}]$$

$$3 = 291 \cdot 13 - 252 \cdot (13 + 2) \quad [\text{Collect terms involving 252}]$$

$$3 = 291 \cdot 13 - 252 \cdot 15 \quad [\text{Bézout's Identity: } au + bv = d]$$

We have found integers $u = 13$ and $v = -15$ such that $291u + 252v = 3$. **Result:**
 $291(13) + 252(-15) = 3$.

Exercise 2.2

Find all the integer solutions to $15u + 25v = 45$. Find also all natural solutions, if any.

Solution. The given linear Diophantine equation is $15u + 25v = 45$.

Step 1: Find $d = \gcd(15, 25)$ We use the Euclidean Algorithm:

$$25 = 15 \cdot 1 + 10$$

$$15 = 10 \cdot 1 + 5$$

$$10 = 5 \cdot 2 + 0$$

Thus, $d = \gcd(15, 25) = 5$.

Step 2: Check Solvability and Reduce the Equation The equation has integer solutions if and only if d divides $c = 45$. Since $5|45$ (as $45 = 5 \cdot 9$), integer solutions exist. We divide the entire equation by $d = 5$ to simplify it:

$$\frac{15}{5}u + \frac{25}{5}v = \frac{45}{5} \implies 3u + 5v = 9$$

Let $a' = 3$, $b' = 5$, and $c' = 9$. Note that $\gcd(a', b') = \gcd(3, 5) = 1$.

Step 3: Find a Particular Solution to $3u + 5v = 9$ We first find a solution (u_*, v_*) for $3u + 5v = 1$ using the Extended Euclidean Algorithm on $(3, 5)$:

$$5 = 3 \cdot 1 + 2$$

$$3 = 2 \cdot 1 + 1$$

$$2 = 1 \cdot 2 + 0$$

Working backward to express 1:

$$1 = 3 - 2 \cdot 1$$

$$1 = 3 - (5 - 3 \cdot 1) \cdot 1$$

$$1 = 3 - 5 + 3 \cdot 1 = 3(2) + 5(-1)$$

So, a solution to $3u + 5v = 1$ is $u_* = 2$ and $v_* = -1$.

Since our equation is $3u + 5v = 9$, we multiply the solution by $c' = 9$:

$$u_0 = u_* \cdot 9 = 2 \cdot 9 = 18$$

$$v_0 = v_* \cdot 9 = -1 \cdot 9 = -9$$

Thus, $(u_0, v_0) = (18, -9)$ is a particular integer solution.

Step 4: Find all Integer Solutions Since $\gcd(3, 5) = 1$, all integer solutions (u, v) to the reduced equation $3u + 5v = 9$ are given by:

$$u = u_0 + kb' \quad \text{and} \quad v = v_0 - ka' \quad \text{for any } k \in \mathbb{Z}$$

$$u = 18 + 5k \quad \text{and} \quad v = -9 - 3k \quad \text{for any } k \in \mathbb{Z} \quad [\text{where } a' = 3, b' = 5]$$

Step 5: Find all Natural Solutions ($\mathbb{N} \cup \{0\}$) We need to find $k \in \mathbb{Z}$ such that $u \geq 0$ and $v \geq 0$:

$$u \geq 0 \implies 18 + 5k \geq 0 \implies 5k \geq -18 \implies k \geq -18/5 = -3.6$$

$$v \geq 0 \implies -9 - 3k \geq 0 \implies -9 \geq 3k \implies k \leq -9/3 = -3$$

Since k must be an integer, we have $-3.6 \leq k \leq -3$. The only integer satisfying this condition is $k = -3$.

Substituting $k = -3$ into the general solution:

$$u = 18 + 5(-3) = 18 - 15 = 3$$

$$v = -9 - 3(-3) = -9 + 9 = 0$$

Result:

- ****All Integer Solutions:**** $u = 18 + 5k$, $v = -9 - 3k$, where $k \in \mathbb{Z}$.
- ****All Natural Solutions:**** The unique solution is $(u, v) = (3, 0)$.

Exercise 2.10

Let p be a prime number. Without using the fundamental theorem of arithmetic, prove that if $m \in \mathbb{Z}$ and $p \nmid m$, then p and m are coprime.

Proof. Let $d = \gcd(p, m)$. By the definition of the greatest common divisor, d must be a positive common divisor of p and m .

Since p is a **prime number**, its only positive divisors are 1 and p . Therefore, the greatest common divisor d must be one of these two values: $d = 1$ or $d = p$.

We proceed by **contradiction**. Assume that $d \neq 1$, so we must have $d = p$. If $d = p$, then p is a common divisor of p and m .

- Since d divides p , $p|p$ (which is true).
- Since d divides m , $p|m$.

However, the initial hypothesis is that $p \nmid m$ (p does not divide m). The conclusion $p|m$ contradicts the given hypothesis $p \nmid m$.

Therefore, the assumption $d = p$ must be false. The only remaining possibility is $d = 1$. Thus, $\gcd(p, m) = 1$, meaning p and m are coprime. $\square$

Week 3

Exercise 3.1

Find all the solutions to $3x \equiv 8 \pmod{11}$.

Solution. The linear congruence $ax \equiv b \pmod{m}$ is equivalent to the linear Diophantine equation $ax + mv = b$. Here, $3x + 11v = 8$.

Step 1: Check Solvability We find $d = \gcd(a, m) = \gcd(3, 11)$. Since 11 is a prime number and 3 is not a multiple of 11, $\gcd(3, 11) = 1$.

$$\gcd(3, 11) = 1$$

Since $d = 1$ divides $b = 8$, a unique solution modulo $m = 11$ exists [Prop 7.1 & 7.2: Solvability and Number of Solutions].

Step 2: Find the Modular Inverse of 3 (mod 11) We use the Extended Euclidean Algorithm to find u and v such that $3u + 11v = 1$.

$$11 = 3 \cdot 3 + 2$$

$$3 = 2 \cdot 1 + 1$$

$$2 = 1 \cdot 2 + 0$$

Working backward for Bézout's identity:

$$1 = 3 - 2 \cdot 1$$

$$1 = 3 - (11 - 3 \cdot 3) \cdot 1 \quad [\text{Substitute } 2 = 11 - 3 \cdot 3]$$

$$1 = 3 - 11 + 3 \cdot 3$$

$$1 = 3 \cdot (1 + 3) - 11 \cdot 1$$

$$1 = 3 \cdot 4 - 11 \cdot 1$$

Thus, $3(4) \equiv 1 \pmod{11}$. The modular inverse is $a^{-1} = 4$.

Step 3: Solve the Congruence Multiply the original congruence by the inverse 4:

$$3x \equiv 8 \pmod{11}$$

$$4 \cdot (3x) \equiv 4 \cdot 8 \pmod{11}$$

$$12x \equiv 32 \pmod{11}$$

Since $12 \equiv 1 \pmod{11}$ and $32 = 2 \cdot 11 + 10$, so $32 \equiv 10 \pmod{11}$:

$$x \equiv 10 \pmod{11}$$

Since $10 \equiv -1 \pmod{11}$, the unique solution modulo 11 is $x \equiv 10 \pmod{11}$.

Step 4: All Integer Solutions The set of all integer solutions is $x = 10 + 11k$, for $k \in \mathbb{Z}$. **Result:** $x \equiv 10 \pmod{11}$.

Exercise 3.2

Find all the solutions to $91x \equiv 3 \pmod{121}$.

Solution. The linear congruence is $91x \equiv 3 \pmod{121}$, equivalent to $91x + 121v = 3$.

Step 1: Check Solvability We find $d = \gcd(91, 121)$ using the Euclidean Algorithm:

$$121 = 91 \cdot 1 + 30$$

$$91 = 30 \cdot 3 + 1$$

$$30 = 1 \cdot 30 + 0$$

Thus, $d = \gcd(91, 121) = 1$. Since $d = 1$ divides $b = 3$, a unique solution modulo 121 exists [Prop 7.1 & 7.2].

Step 2: Find the Modular Inverse of 91 (mod 121) We use the Extended Euclidean Algorithm to find u such that $91u \equiv 1 \pmod{121}$:

$$1 = 91 - 30 \cdot 3 \quad [\text{From the second equation}]$$

$$1 = 91 - (121 - 91 \cdot 1) \cdot 3 \quad [\text{Substitute } 30 = 121 - 91 \cdot 1]$$

$$1 = 91 - 121 \cdot 3 + 91 \cdot 3$$

$$1 = 91 \cdot (1 + 3) - 121 \cdot 3$$

$$1 = 91 \cdot 4 - 121 \cdot 3$$

Thus, $91(4) \equiv 1 \pmod{121}$. The modular inverse is $a^{-1} = 4$.

Step 3: Solve the Congruence Multiply the original congruence by the inverse 4:

$$91x \equiv 3 \pmod{121}$$

$$4 \cdot (91x) \equiv 4 \cdot 3 \pmod{121}$$

$$(91 \cdot 4)x \equiv 12 \pmod{121}$$

Since $91 \cdot 4 \equiv 1 \pmod{121}$:

$$x \equiv 12 \pmod{121}$$

Result: $x \equiv 12 \pmod{121}$.

Exercise 3.3

Solve $83x \equiv 2 \pmod{100}$.

Solution. The linear congruence is $83x \equiv 2 \pmod{100}$, equivalent to $83x + 100v = 2$.

Step 1: Check Solvability We find $d = \gcd(83, 100)$:

$$100 = 83 \cdot 1 + 17$$

$$83 = 17 \cdot 4 + 15$$

$$17 = 15 \cdot 1 + 2$$

$$15 = 2 \cdot 7 + 1$$

$$2 = 1 \cdot 2 + 0$$

Thus, $d = \gcd(83, 100) = 1$. Since $d = 1$ divides $b = 2$, a unique solution modulo 100 exists.

Step 2: Find the Modular Inverse of 83 (mod 100) We use the Extended Euclidean Algorithm to express 1 as a linear combination of 83 and 100:

$$\begin{aligned}
 1 &= 15 - 2 \cdot 7 \\
 1 &= 15 - (17 - 15 \cdot 1) \cdot 7 && [\text{Substitute } 2 = 17 - 15 \cdot 1] \\
 1 &= 15 - 17 \cdot 7 + 15 \cdot 7 \\
 1 &= 15 \cdot 8 - 17 \cdot 7 \\
 1 &= (83 - 17 \cdot 4) \cdot 8 - 17 \cdot 7 && [\text{Substitute } 15 = 83 - 17 \cdot 4] \\
 1 &= 83 \cdot 8 - 17 \cdot 32 - 17 \cdot 7 \\
 1 &= 83 \cdot 8 - 17 \cdot 39 \\
 1 &= 83 \cdot 8 - (100 - 83 \cdot 1) \cdot 39 && [\text{Substitute } 17 = 100 - 83 \cdot 1] \\
 1 &= 83 \cdot 8 - 100 \cdot 39 + 83 \cdot 39 \\
 1 &= 83 \cdot (8 + 39) - 100 \cdot 39 \\
 1 &= 83 \cdot 47 - 100 \cdot 39
 \end{aligned}$$

Thus, $83(47) \equiv 1 \pmod{100}$. The inverse is $a^{-1} = 47$.

Step 3: Solve the Congruence Multiply the original congruence by the inverse 47:

$$\begin{aligned}
 83x &\equiv 2 \pmod{100} \\
 47 \cdot (83x) &\equiv 47 \cdot 2 \pmod{100} \\
 (83 \cdot 47)x &\equiv 94 \pmod{100} \\
 x &\equiv 94 \pmod{100}
 \end{aligned}$$

The solution in $\mathbb{Z}/100\mathbb{Z}$ is the residue class $[94]_{100}$. **Result:** $x \equiv 94 \pmod{100}$.

Exercise 3.4

Find $5327 \cdot 6185 \cdot 7139 \cdot 2187 \cdot 5219 \cdot 1873 \pmod{8157}$. (Hint: After each multiplication, reduce modulo 8157 before doing the next multiplication.)

Solution. We perform the chained multiplication, reducing the product modulo $m = 8157$ at each step to manage the size of the numbers.

Step 1: First multiplication

$$\begin{aligned}
 5327 \cdot 6185 &= 32950195 \\
 32950195 \div 8157 &= 4039.515263... \\
 r_1 &= 32950195 - 8157 \cdot 4039 = 32950195 - 32950143 = 52 \\
 5327 \cdot 6185 &\equiv 52 \pmod{8157}
 \end{aligned}$$

Step 2: Multiply by 7139

$$\begin{aligned}
 52 \cdot 7139 &= 371228 \\
 371228 \div 8157 &= 45.515263... \\
 r_2 &= 371228 - 8157 \cdot 45 = 371228 - 367065 = 4163 \\
 52 \cdot 7139 &\equiv 4163 \pmod{8157}
 \end{aligned}$$

Step 3: Multiply by 2187

$$4163 \cdot 2187 = 9104081$$

$$9104081 \div 8157 = 1116.096\dots$$

$$r_3 = 9104081 - 8157 \cdot 1116 = 9104081 - 9103812 = 269$$

$$4163 \cdot 2187 \equiv 269 \pmod{8157}$$

Step 4: Multiply by 5219

$$269 \cdot 5219 = 1403611$$

$$1403611 \div 8157 = 172.079\dots$$

$$r_4 = 1403611 - 8157 \cdot 172 = 1403611 - 1402904 = 707$$

$$269 \cdot 5219 \equiv 707 \pmod{8157}$$

Step 5: Multiply by 1873

$$707 \cdot 1873 = 1324151$$

$$1324151 \div 8157 = 162.338\dots$$

$$r_5 = 1324151 - 8157 \cdot 162 = 1324151 - 1321434 = 2717$$

$$707 \cdot 1873 \equiv 2717 \pmod{8157}$$

Result: $5327 \cdot 6185 \cdot 7139 \cdot 2187 \cdot 5219 \cdot 1873 \equiv 2717 \pmod{8157}$.

Exercise 3.5

Find $373^6 \pmod{581}$ (Hint: as above).

Solution. We use the ****Fast Powering Algorithm**** by successive squaring and reduction modulo $m = 581$. Since $6 = (110)_2 = 2^2 + 2^1$, we need to compute 373^2 , 373^4 , and then $373^6 = 373^4 \cdot 373^2$.

Step 1: Compute $373^2 \pmod{581}$

$$373^2 = 139129$$

$$139129 \div 581 = 239.4647\dots$$

$$r_1 = 139129 - 581 \cdot 239 = 139129 - 138959 = 170$$

$$373^2 \equiv 170 \pmod{581}$$

Step 2: Compute $373^4 \pmod{581}$

$$373^4 \equiv (373^2)^2 \equiv 170^2 \pmod{581}$$

$$170^2 = 28900$$

$$28900 \div 581 = 49.7418\dots$$

$$r_2 = 28900 - 581 \cdot 49 = 28900 - 28469 = 431$$

$$373^4 \equiv 431 \pmod{581}$$

Step 3: Compute $373^6 \pmod{581}$

$$373^6 \equiv 373^4 \cdot 373^2 \equiv 431 \cdot 170 \pmod{581}$$

$$431 \cdot 170 = 73270$$

$$73270 \div 581 = 126.1101\dots$$

$$r_3 = 73270 - 581 \cdot 126 = 73270 - 73150 = 120$$

$$373^6 \equiv 120 \pmod{581}$$

Result: $373^6 \equiv 120 \pmod{581}$.

Exercise 3.6

Prove that a is invertible in $\mathbb{Z}/m\mathbb{Z}$ if and only if $\gcd(a, m) = 1$.

Proof. We prove the biconditional statement by demonstrating both directions.

($\implies$) **If $[a]$ is invertible, then $\gcd(a, m) = 1$** If the residue class $[a]$ is invertible in $\mathbb{Z}/m\mathbb{Z}$, then there exists an integer $u \in \mathbb{Z}$ (the inverse) such that:

$$au \equiv 1 \pmod{m}$$

By the definition of modular congruence, this means that m divides the difference $(au - 1)$.

$$m \mid (au - 1)$$

This implies the existence of an integer $v \in \mathbb{Z}$ such that:

$$au - 1 = -mv$$

Rearranging the terms gives the linear Diophantine equation:

$$au + mv = 1$$

Let $d = \gcd(a, m)$. By the property of divisibility, since d divides a and d divides m , d must divide any linear combination of a and m :

$$d \mid (au + mv) \implies d \mid 1$$

Since d is a positive integer, the only value satisfying $d \mid 1$ is $d = 1$. Thus, $\gcd(a, m) = 1$.

($\impliedby$) **If $\gcd(a, m) = 1$, then $[a]$ is invertible** Assume $\gcd(a, m) = 1$. By Bézout's Identity, there exist integers $u, v \in \mathbb{Z}$ such that:

$$au + mv = 1$$

Rearranging the terms:

$$au - 1 = -mv$$

By the definition of divisibility, since $-mv$ is a multiple of m , m divides $(au - 1)$:

$$m \mid (au - 1)$$

By the definition of modular congruence, this means:

$$au \equiv 1 \pmod{m}$$

This shows that u is the multiplicative inverse of a modulo m , meaning the residue class $[a]$ is invertible in $\mathbb{Z}/m\mathbb{Z}$. $\square$

Exercise 3.7

Let $m \geq 1$ be an integer and suppose that $a_1 \equiv a_2 \pmod{m}$ and $b_1 \equiv b_2 \pmod{m}$. Prove that $a_1 + b_1 \equiv a_2 + b_2 \pmod{m}$ and $a_1 \cdot b_1 \equiv a_2 \cdot b_2 \pmod{m}$.

Proof. The premise $a_1 \equiv a_2 \pmod{m}$ means $m \mid (a_1 - a_2)$, so $a_1 - a_2 = k_1 m$ for some $k_1 \in \mathbb{Z}$. Similarly, $b_1 \equiv b_2 \pmod{m}$ means $m \mid (b_1 - b_2)$, so $b_1 - b_2 = k_2 m$ for some $k_2 \in \mathbb{Z}$.

Proof for Addition ($a_1 + b_1 \equiv a_2 + b_2 \pmod{m}$) We want to show $m \mid ((a_1 + b_1) - (a_2 + b_2))$.

$$(a_1 + b_1) - (a_2 + b_2) = (a_1 - a_2) + (b_1 - b_2) \quad [\text{Rearranging terms}]$$

Substituting the expressions from the premise:

$$(a_1 - a_2) + (b_1 - b_2) = k_1 m + k_2 m = m(k_1 + k_2)$$

Since $k_1 + k_2 \in \mathbb{Z}$, the difference is a multiple of m . Thus, $m \mid ((a_1 + b_1) - (a_2 + b_2))$, which means:

$$a_1 + b_1 \equiv a_2 + b_2 \pmod{m}$$

Proof for Multiplication ($a_1 \cdot b_1 \equiv a_2 \cdot b_2 \pmod{m}$) We use the definition of congruence to write a_1 and b_1 in terms of a_2 and b_2 :

$$a_1 = a_2 + k_1 m \quad \text{and} \quad b_1 = b_2 + k_2 m$$

Now compute the product $a_1 b_1$:

$$\begin{aligned} a_1 b_1 &= (a_2 + k_1 m)(b_2 + k_2 m) \\ a_1 b_1 &= a_2 b_2 + a_2 k_2 m + b_2 k_1 m + k_1 k_2 m^2 \\ a_1 b_1 - a_2 b_2 &= m(a_2 k_2 + b_2 k_1 + k_1 k_2 m) \end{aligned}$$

Let $K = a_2 k_2 + b_2 k_1 + k_1 k_2 m$. Since $K \in \mathbb{Z}$, the difference $a_1 b_1 - a_2 b_2$ is a multiple of m . Thus, $m \mid (a_1 b_1 - a_2 b_2)$, which means:

$$a_1 \cdot b_1 \equiv a_2 \cdot b_2 \pmod{m}$$

□

Exercise 3.8

1. Suppose m is odd. Find 2^{-1} in $\mathbb{Z}/m\mathbb{Z}$.
2. More generally, suppose $m \equiv 1 \pmod{b}$ for some $b \geq 2$. Find b^{-1} in $\mathbb{Z}/m\mathbb{Z}$.

Solution. 1. **Find $2^{-1} \pmod{m}$ where m is odd** Since m is odd, $\gcd(2, m) = 1$, so the inverse 2^{-1} exists [Thm 6.2: Characterization of Invertibility]. The inverse $x = 2^{-1}$ is the solution to the congruence $2x \equiv 1 \pmod{m}$, or the Diophantine equation $2x + mv = 1$. We need $2x = 1 - mv$. Since m is odd, we choose an odd multiple of m , km , such that $km + 1$ is even. The simplest choice is $k = 1$. Let $v = -1$ (i.e., we choose the multiple m):

$$2x = 1 + m \quad [\text{Since } m \equiv -1 \pmod{2}, 1 + m \text{ is even}]$$

$$x = \frac{m+1}{2}$$

Since m is odd, $m+1$ is even, and x is an integer. **Result:** $2^{-1} \equiv \frac{m+1}{2} \pmod{m}$.

2. **Find** $b^{-1} \pmod{m}$ **where** $m \equiv 1 \pmod{b}$ The condition $m \equiv 1 \pmod{b}$ means that b divides $(m-1)$, so $m-1 = bq$ for some integer q .

$$m = bq + 1$$

From this equation, we can write 1 as a linear combination of b and m :

$$b(-q) + m(1) = 1 \quad [\text{Bézout's Identity}]$$

This equation shows that $-q$ is the inverse of b modulo m :

$$b(-q) \equiv 1 \pmod{m}$$

Since $m = bq + 1$, we have $q = \frac{m-1}{b}$. The inverse is $-q = -\frac{m-1}{b}$. To find the positive representative, we add m :

$$b^{-1} \equiv -q + m \equiv -q \pmod{m}$$

Result: $b^{-1} \equiv -q \equiv -\frac{m-1}{b} \pmod{m}$. The positive residue is $m - \frac{m-1}{b}$.

Exercise 3.9

Let m be an odd integer and let a be any integer. Prove that $2m + a^2$ can never be the square of an integer. (Hint: If a natural is a square, what are its possible values modulo 4?)

Proof. We use modular arithmetic modulo 4.

Step 1: Determine the possible values of a perfect square modulo 4 Let $n \in \mathbb{Z}$. Since any integer n is either even ($n \equiv 0, 2 \pmod{4}$) or odd ($n \equiv 1, 3 \pmod{4}$), we examine $n^2 \pmod{4}$:

- If n is even, $n \equiv 0 \pmod{2}$, so $n = 2k$. $n^2 = 4k^2 \equiv 0 \pmod{4}$.
- If n is odd, $n \equiv 1 \pmod{2}$, so $n = 2k + 1$. $n^2 = 4k^2 + 4k + 1 \equiv 1 \pmod{4}$.

Therefore, any perfect square n^2 must be congruent to 0 or 1 modulo 4.

Step 2: Determine the possible values of $2m + a^2$ modulo 4 The term $2m + a^2$ has two parts:

- **$2m \pmod{4}$:** Since m is odd, $m \equiv 1 \pmod{2}$, so m is of the form $2k + 1$.

$$2m = 2(2k + 1) = 4k + 2 \equiv 2 \pmod{4}$$

- **$a^2 \pmod{4}$:** As shown in Step 1, $a^2 \equiv 0$ or $1 \pmod{4}$.

Now, we find the possible values for the sum $2m + a^2 \pmod{4}$:

- Case 1: $a^2 \equiv 0 \pmod{4}$.

$$2m + a^2 \equiv 2 + 0 \equiv 2 \pmod{4}$$

- Case 2: $a^2 \equiv 1 \pmod{4}$.

$$2m + a^2 \equiv 2 + 1 \equiv 3 \pmod{4}$$

Thus, $2m + a^2$ is congruent to 2 or 3 modulo 4.

Step 3: Conclusion Since $2m + a^2$ is congruent to 2 or 3 (mod 4), and no perfect square is congruent to 2 or 3 (mod 4) (from Step 1), $2m + a^2$ cannot be a perfect square. $\square$

Exercise 3.12

Find all the solutions (if any) to $4x \equiv 6 \pmod{30}$.

Solution. The linear congruence is $4x \equiv 6 \pmod{30}$, equivalent to the linear Diophantine equation $4x + 30v = 6$.

Step 1: Check Solvability We find $d = \gcd(4, 30)$:

$$30 = 4 \cdot 7 + 2$$

$$4 = 2 \cdot 2 + 0$$

Thus, $d = \gcd(4, 30) = 2$.

The equation is solvable if and only if d divides $b = 6$. Since $2|6$, solutions exist. The number of distinct solutions modulo $m = 30$ is $d = 2$ [Prop 7.2: Number of Solutions].

Step 2: Reduce the Congruence We divide the entire congruence by $d = 2$:

$$\frac{4}{2}x \equiv \frac{6}{2} \pmod{\frac{30}{2}}$$

$$2x \equiv 3 \pmod{15}$$

This reduced congruence $2x \equiv 3 \pmod{15}$ has a unique solution modulo 15, since $\gcd(2, 15) = 1$.

Step 3: Find the Modular Inverse of 2 (mod 15) We need $2^{-1} \pmod{15}$. We look for a number u such that $2u \equiv 1 \pmod{15}$.

$$2u = 1 + 15k$$

By inspection (or EEA): $2 \cdot 8 = 16 \equiv 1 \pmod{15}$. The inverse is $u = 8$.

Step 4: Solve the Reduced Congruence Multiply the reduced congruence by the inverse 8:

$$8 \cdot (2x) \equiv 8 \cdot 3 \pmod{15}$$

$$16x \equiv 24 \pmod{15}$$

Since $16 \equiv 1 \pmod{15}$ and $24 = 1 \cdot 15 + 9$, so $24 \equiv 9 \pmod{15}$:

$$x \equiv 9 \pmod{15}$$

Step 5: Find all Solutions Modulo 30 The solution is $x = 9 + 15k$ for $k \in \mathbb{Z}$. We need the solutions in the range $0 \leq x < 30$:

- For $k = 0$: $x = 9 + 15(0) = 9$
- For $k = 1$: $x = 9 + 15(1) = 24$

Result: The two distinct solutions modulo 30 are $x \equiv 9 \pmod{30}$ and $x \equiv 24 \pmod{30}$.

Week 4

Exercise 4.1

Find all the solutions to the system of congruences:

$$\begin{cases} x \equiv 137 \pmod{423} \\ x \equiv 87 \pmod{191} \end{cases}$$

Use the method by hand (substitution).

Solution. The system has the form $x \equiv a_1 \pmod{m_1}$ and $x \equiv a_2 \pmod{m_2}$, with $m_1 = 423$, $m_2 = 191$, $a_1 = 137$, and $a_2 = 87$.

Step 1: Check Solvability and Coprimality We first check if $\gcd(423, 191) = 1$ using the Euclidean Algorithm:

$$\begin{aligned} 423 &= 191 \cdot 2 + 41 \\ 191 &= 41 \cdot 4 + 27 \\ 41 &= 27 \cdot 1 + 14 \\ 27 &= 14 \cdot 1 + 13 \\ 14 &= 13 \cdot 1 + 1 \\ 13 &= 1 \cdot 13 + 0 \end{aligned}$$

Since the last non-zero remainder is 1, $\gcd(423, 191) = 1$. A unique solution modulo $M = 423 \cdot 191$ exists [Thm 8.1: Chinese Remainder Theorem].

Step 2: Substitution and Reduction From the first congruence, x must be of the form:

$$x = 137 + 423k \quad \text{for some } k \in \mathbb{Z} \quad [\text{Def: Congruence}]$$

Substitute this into the second congruence:

$$\begin{aligned} 137 + 423k &\equiv 87 \pmod{191} \\ 423k &\equiv 87 - 137 \pmod{191} \\ 423k &\equiv -50 \pmod{191} \end{aligned}$$

To simplify, we reduce the coefficient 423 modulo 191 [Prop 6.1: Compatibility, for multiplication]. Using the first step of the Euclidean Algorithm:

$$423 = 191 \cdot 2 + 41 \implies 423 \equiv 41 \pmod{191}$$

Also, $-50 \equiv -50 + 191 \equiv 141 \pmod{191}$. The linear congruence for k becomes:

$$41k \equiv 141 \pmod{191} \quad [\text{Reduced congruence for } k]$$

Step 3: Solve for k We need the inverse of 41 (mod 191). We use the intermediate steps from the Euclidean Algorithm in Step 1 to apply the Extended Euclidean Algorithm:

$$\begin{aligned}
 1 &= 14 - 13 \cdot 1 \\
 1 &= 14 - (27 - 14 \cdot 1) \cdot 1 && [\text{Substitute } 13 = 27 - 14 \cdot 1] \\
 1 &= 14 \cdot 2 - 27 \cdot 1 \\
 1 &= (41 - 27 \cdot 1) \cdot 2 - 27 \cdot 1 && [\text{Substitute } 14 = 41 - 27 \cdot 1] \\
 1 &= 41 \cdot 2 - 27 \cdot 3 \\
 1 &= 41 \cdot 2 - (191 - 41 \cdot 4) \cdot 3 && [\text{Substitute } 27 = 191 - 41 \cdot 4] \\
 1 &= 41 \cdot 2 - 191 \cdot 3 + 41 \cdot 12 \\
 1 &= 41 \cdot 14 - 191 \cdot 3
 \end{aligned}$$

Thus, $41(14) \equiv 1 \pmod{191}$. The inverse is $41^{-1} \equiv 14 \pmod{191}$.

Now, multiply the congruence for k by the inverse 14:

$$14 \cdot (41k) \equiv 14 \cdot 141 \pmod{191}$$

$$k \equiv 1974 \pmod{191}$$

We reduce 1974 modulo 191:

$$1974 = 191 \cdot 10 + 64$$

$$k \equiv 64 \pmod{191} \quad [\text{Solution for } k]$$

This means $k = 64 + 191l$ for some $l \in \mathbb{Z}$.

Step 4: Final Solution for x Substitute the expression for k back into $x = 137 + 423k$:

$$\begin{aligned}
 x &= 137 + 423(64 + 191l) \\
 x &= 137 + 423 \cdot 64 + 423 \cdot 191l \\
 x &= 137 + 27072 + (423 \cdot 191)l \\
 x &= 27209 + 80793l
 \end{aligned}$$

The modulus for the solution is $M = 423 \cdot 191 = 80793$. We reduce the particular solution $x_0 = 27209$ modulo 80793. Since $27209 < 80793$, the solution is $x \equiv 27209 \pmod{80793}$.

Result: $x \equiv 27209 \pmod{80793}$.

—

Exercise 4.2 (Smallest Secret Number)

An adventurer has the following conditions for a secret number x :

- x , when divided by 3, leaves a remainder of 2.
- x , when divided by 5, leaves a remainder of 3.
- x , when divided by 7, leaves a remainder of 4.

Find the smallest secret number that satisfies all three conditions.

Solution. This system of congruences is:

$$\begin{cases} x \equiv 2 \pmod{3} & (a_1 = 2, m_1 = 3) \\ x \equiv 3 \pmod{5} & (a_2 = 3, m_2 = 5) \\ x \equiv 4 \pmod{7} & (a_3 = 4, m_3 = 7) \end{cases}$$

Since 3, 5, 7 are pairwise coprime, a unique solution exists modulo $M = 3 \cdot 5 \cdot 7 = 105$ [Thm 8.1: Chinese Remainder Theorem]. We use the constructive method.

Step 1: Compute M and N_i

$$M = m_1 m_2 m_3 = 3 \cdot 5 \cdot 7 = 105$$

$$N_1 = M/m_1 = 105/3 = 35$$

$$N_2 = M/m_2 = 105/5 = 21$$

$$N_3 = M/m_3 = 105/7 = 15$$

Step 2: Find the Modular Inverses y_i (Bézout coefficients) We solve $N_i y_i \equiv 1 \pmod{m_i}$ for y_i .

- $i = 1$: $35y_1 \equiv 1 \pmod{3}$. Since $35 = 3 \cdot 11 + 2$, $35 \equiv 2 \pmod{3}$.

$$2y_1 \equiv 1 \pmod{3}$$

Since $2 \cdot 2 = 4 \equiv 1 \pmod{3}$, the inverse is $y_1 = 2$.

- $i = 2$: $21y_2 \equiv 1 \pmod{5}$. Since $21 = 5 \cdot 4 + 1$, $21 \equiv 1 \pmod{5}$.

$$1y_2 \equiv 1 \pmod{5}$$

The inverse is $y_2 = 1$.

- $i = 3$: $15y_3 \equiv 1 \pmod{7}$. Since $15 = 7 \cdot 2 + 1$, $15 \equiv 1 \pmod{7}$.

$$1y_3 \equiv 1 \pmod{7}$$

The inverse is $y_3 = 1$.

Step 3: Construct the Solution x_0 The particular solution is constructed as $x_0 = \sum_{i=1}^3 a_i N_i y_i$ [Thm 8.1 Proof: Existence (Construction)].

$$x_0 = a_1 N_1 y_1 + a_2 N_2 y_2 + a_3 N_3 y_3$$

$$x_0 = (2 \cdot 35 \cdot 2) + (3 \cdot 21 \cdot 1) + (4 \cdot 15 \cdot 1)$$

$$x_0 = 140 + 63 + 60 = 263 \quad [\text{Intermediate sum}]$$

Step 4: Find the Smallest Positive Solution The solution x is unique modulo $M = 105$. We reduce $x_0 = 263$ modulo 105:

$$263 = 105 \cdot 2 + 53$$

$$x \equiv 53 \pmod{105} \quad [\text{Solution modulo } M]$$

The smallest secret number is the smallest positive residue, which is 53. **Result:** The smallest secret number is $x = 53$.

—

Exercise 4.6

Find all the solutions to the system of congruences:

$$\begin{cases} x \equiv 5 \pmod{9} \\ x \equiv 6 \pmod{10} \\ x \equiv 7 \pmod{11} \end{cases}$$

Use the constructive method of the proof of the Theorem (CRT).

Solution. The system is defined by moduli $m_i = \{9, 10, 11\}$ and residues $a_i = \{5, 6, 7\}$. Since $\gcd(9, 10) = 1$, $\gcd(9, 11) = 1$, and $\gcd(10, 11) = 1$, the moduli are pairwise coprime, and a unique solution exists modulo $M = 9 \cdot 10 \cdot 11$ [Thm 8.1: Chinese Remainder Theorem].

Step 1: Compute M and N_i

$$M = 9 \cdot 10 \cdot 11 = 990$$

$$N_1 = M/m_1 = 990/9 = 110$$

$$N_2 = M/m_2 = 990/10 = 99$$

$$N_3 = M/m_3 = 990/11 = 90 \quad [\text{The } N_i \text{ values are } \{110, 99, 90\}]$$

Step 2: Find the Modular Inverses y_i We solve $N_i y_i \equiv 1 \pmod{m_i}$ for y_i .

- $i = 1$: $110y_1 \equiv 1 \pmod{9}$. Since $110 = 9 \cdot 12 + 2$, $110 \equiv 2 \pmod{9}$.

$$2y_1 \equiv 1 \pmod{9}$$

Since $2 \cdot 5 = 10 \equiv 1 \pmod{9}$, the inverse is $y_1 = 5$.

- $i = 2$: $99y_2 \equiv 1 \pmod{10}$. Since $99 = 10 \cdot 9 + 9$, $99 \equiv 9 \pmod{10}$.

$$9y_2 \equiv 1 \pmod{10}$$

Since $9 \equiv -1 \pmod{10}$, and $(-1)(-1) = 1$, the inverse is $y_2 = -1 \equiv 9 \pmod{10}$.

- $i = 3$: $90y_3 \equiv 1 \pmod{11}$. Since $90 = 11 \cdot 8 + 2$, $90 \equiv 2 \pmod{11}$.

$$2y_3 \equiv 1 \pmod{11}$$

Since $2 \cdot 6 = 12 \equiv 1 \pmod{11}$, the inverse is $y_3 = 6$.

Step 3: Construct the Solution x_0

$$x_0 = \sum_{i=1}^3 a_i N_i y_i$$

$$x_0 = (5 \cdot 110 \cdot 5) + (6 \cdot 99 \cdot 9) + (7 \cdot 90 \cdot 6)$$

$$x_0 = 2750 + 5346 + 3780 = 11876 \quad [\text{Intermediate sum}]$$

Step 4: Find the Smallest Positive Solution The solution x is unique modulo $M = 990$. We reduce $x_0 = 11876$ modulo 990:

$$11876 \div 990 \approx 12 \quad [\text{Quotient is } 12]$$

$$r = 11876 - 990 \cdot 12 = 11876 - 11880 = -4$$

$$x \equiv -4 \equiv -4 + 990 \equiv 986 \pmod{990} \quad [\text{Smallest positive solution}]$$

Result: The set of all solutions is $x \equiv 986 \pmod{990}$.

Week 5

Exercise 5.4

What is the remainder of the division of 221^{654} by 109 (109 is prime)?

Solution. We want to compute $221^{654} \pmod{109}$. The modulus $p = 109$ is prime.

Step 1: Reduce the Base Modulo 109

$$221 = 2 \cdot 109 + 3 \implies 221 \equiv 3 \pmod{109}$$

The expression becomes:

$$221^{654} \equiv 3^{654} \pmod{109}$$

Step 2: Reduce the Exponent Modulo $\phi(109) = 108$ Since 109 is prime and $109 \nmid 3$, we can use Fermat's Little Theorem (FLT): $3^{108} \equiv 1 \pmod{109}$. We reduce the exponent 654 modulo $p - 1 = 108$:

$$654 = 108 \cdot 6 + 6$$

$$654 \equiv 6 \pmod{108}$$

Step 3: Compute the Final Result We apply the modular exponentiation property [Cor 11.1: Modular Exponentiation]:

$$3^{654} \equiv 3^{108 \cdot 6 + 6} \equiv (3^{108})^6 \cdot 3^6 \pmod{109}$$

$$3^{654} \equiv (1)^6 \cdot 3^6 \equiv 3^6 \pmod{109}$$

Now we compute 3^6 :

$$3^6 = 729$$

Finally, we reduce 729 modulo 109:

$$729 = 6 \cdot 109 + 75 \implies 729 \equiv 75 \pmod{109}$$

Result: The remainder of the division of 221^{654} by 109 is 75.

—

Exercise 5.6

Compute $7^{50} \pmod{47}$.

Solution. We compute $7^{50} \pmod{47}$. Since 47 is prime, we use the Fast Powering Algorithm (FPA) combined with reduction techniques.

Step 1: Binary Representation of the Exponent $A = 50$

$$50_{10} = (110010)_2$$

Thus, $50 = 2^5 + 2^4 + 2^1$.

Step 2: Successive Squaring (Powers $7^{2^i} \pmod{47}$) We compute the required powers, reducing modulo 47 at each step.

$$\begin{aligned}
 7^{2^0} &\equiv 7 \pmod{47} \\
 7^{2^1} &\equiv (7^1)^2 = 49 \equiv 2 \pmod{47} & [49 = 1 \cdot 47 + 2] \\
 7^{2^2} &\equiv (2)^2 = 4 \pmod{47} \\
 7^{2^3} &\equiv (4)^2 = 16 \pmod{47} \\
 7^{2^4} &\equiv (16)^2 = 256 \equiv 21 \pmod{47} & [256 = 5 \cdot 47 + 21] \\
 7^{2^5} &\equiv (21)^2 = 441 \equiv 18 \pmod{47} & [441 = 9 \cdot 47 + 18]
 \end{aligned}$$

Step 3: Final Product (Multiplying terms where the bit $A_i = 1$)

$$\begin{aligned}
 7^{50} &\equiv 7^{2^5} \cdot 7^{2^4} \cdot 7^{2^1} \pmod{47} \\
 7^{50} &\equiv 18 \cdot 21 \cdot 2 \pmod{47}
 \end{aligned}$$

We compute the product:

$$\begin{aligned}
 18 \cdot 21 &= 378 \\
 378 \cdot 2 &= 756
 \end{aligned}$$

Reduce the final product modulo 47:

$$756 = 16 \cdot 47 + 4 \implies 756 \equiv 4 \pmod{47}$$

Result: $7^{50} \equiv 4 \pmod{47}$.

—

Exercise 5.7

Find all the solutions to $x^{1383} \equiv 5$ in $\mathbb{F}_{13}$.

Solution. We are solving the congruence $x^{1383} \equiv 5 \pmod{13}$. The modulus $p = 13$ is prime.

Step 1: Simplify the Exponent using FLT Since $p = 13$ is prime, Fermat's Little Theorem states $x^{12} \equiv 1 \pmod{13}$ for all $x \not\equiv 0 \pmod{13}$. We reduce the exponent 1383 modulo $p - 1 = 12$:

$$\begin{aligned}
 1383 &= 12 \cdot 115 + 3 \\
 1383 &\equiv 3 \pmod{12}
 \end{aligned}$$

Step 2: Reduce the Congruence Applying the modular exponentiation property [Cor 11.1: Modular Exponentiation]:

$$x^{1383} \equiv x^{12 \cdot 115 + 3} \equiv (x^{12})^{115} \cdot x^3 \equiv (1)^{115} \cdot x^3 \equiv x^3 \pmod{13}$$

The original equation simplifies to a cubic congruence:

$$x^3 \equiv 5 \pmod{13}$$

Step 3: Test Potential Solutions We test values for $x \in \{0, 1, \dots, 12\}$ to find which ones satisfy $x^3 \equiv 5 \pmod{13}$.

- $x = 7$: $7^3 = 343 \equiv 5 \pmod{13} \implies x \equiv 7$ is a solution.
- $x = 8$: $8^3 = 512 \equiv 5 \pmod{13} \implies x \equiv 8$ is a solution.
- $x = 11$: $11^3 = 1331 \equiv 5 \pmod{13} \implies x \equiv 11$ is a solution.
- Other values are not solutions.

Result: The solutions are $x \in \{7, 8, 11\}$ in $\mathbb{F}_{13}$.

—

Exercise 5.10

Let p be a prime, b an integer, and $p \nmid b$. Show that $\text{ord}_p(b^{-1}) = \text{ord}_p(b)$.

Proof. Let $n = \text{ord}_p(b)$ and $m = \text{ord}_p(b^{-1})$. We aim to show that $n = m$.

Part 1: Show $m|n$ Since $n = \text{ord}_p(b)$, by the definition of the order:

$$b^n \equiv 1 \pmod{p}$$

Multiply both sides by $(b^{-1})^n$:

$$b^n \cdot (b^{-1})^n \equiv 1 \cdot (b^{-1})^n \pmod{p} \quad [\text{Prop 6.1: Compatibility}]$$

$$(b \cdot b^{-1})^n \equiv (b^{-1})^n \pmod{p} \quad [\text{Exponent rules}]$$

Since $b \cdot b^{-1} \equiv 1 \pmod{p}$:

$$1^n \equiv (b^{-1})^n \pmod{p} \implies (b^{-1})^n \equiv 1 \pmod{p}$$

Since $m = \text{ord}_p(b^{-1})$ is the minimum positive integer for which this congruence holds, m must divide n [Prop 12.1: Order Divisibility Property].

$$m|n$$

Part 2: Show $n|m$ The element b is the inverse of b^{-1} , i.e., $b = (b^{-1})^{-1}$ [Cor 11.2: Modular Inverses property]. Since $m = \text{ord}_p(b^{-1})$, we have:

$$(b^{-1})^m \equiv 1 \pmod{p}$$

Multiplying both sides by b^m :

$$(b^{-1})^m \cdot b^m \equiv 1 \cdot b^m \pmod{p} \implies 1 \equiv b^m \pmod{p}$$

Since $n = \text{ord}_p(b)$ is the minimum positive integer for which this congruence holds, n must divide m [Prop 12.1: Order Divisibility Property].

$$n|m$$

Conclusion Since $n|m$ and $m|n$, and both n and m are positive integers (by definition of the order), it must be that $n = m$. Thus, $\text{ord}_p(b^{-1}) = \text{ord}_p(b)$. $\square$

Week 6

Exercise 6.1

Find the order of 3 modulo 127.

Solution. The modulus $p = 127$ is a prime number. The order of any element $a \in \mathbb{F}_p^*$ must divide $\phi(p) = p - 1$ [Prop 12.3: Order Divisibility Property, consequence of FLT].

$$\phi(127) = 127 - 1 = 126 \quad [\text{Thm 9.2: Totient of a Prime Number}]$$

Step 1: Prime Factorization of the Order We find the prime factorization of $p - 1 = 126$:

$$126 = 2 \cdot 63 = 2 \cdot 3^2 \cdot 7 \quad [\text{Prime factors are } 2, 3, 7]$$

The order of 3, $\text{ord}_{127}(3)$, must be a divisor of 126. To check if the order is 126, we use the effective test for primitive roots [Lemma 12.1].

Step 2: Effective Test for Order The order $n = \text{ord}_{127}(3)$ is 126 if and only if $3^{126/q_i} \not\equiv 1 \pmod{127}$ for all prime factors $q_i \in \{2, 3, 7\}$ of 126.

We compute the test powers:

1. $q_1 = 2$: Exponent $E_1 = \frac{126}{2} = 63$ We compute $3^{63} \pmod{127}$ using FPA (since $63 = (111111)_2$):

$$3^{63} \equiv 126 \equiv -1 \pmod{127} \quad [\text{Computed value } 126 \not\equiv 1]$$

2. $q_2 = 3$: Exponent $E_2 = \frac{126}{3} = 42$ We compute $3^{42} \pmod{127}$:

$$3^{42} \equiv 107 \pmod{127} \quad [\text{Computed value } 107 \not\equiv 1]$$

3. $q_3 = 7$: Exponent $E_3 = \frac{126}{7} = 18$ We compute $3^{18} \pmod{127}$:

$$3^{18} \equiv 4 \pmod{127} \quad [\text{Computed value } 4 \not\equiv 1]$$

Step 3: Conclusion Since $3^{E_i} \not\equiv 1 \pmod{127}$ for all $E_i = 126/q_i$, the order of 3 is not a proper divisor of 126. Therefore, the order must be the maximum possible value [Lemma 12.1]:

$$\text{ord}_{127}(3) = 126 \quad [\text{Max order } p - 1]$$

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Exercise 6.3

Given that 2 is a primitive root $\pmod{101}$, find $x \in \mathbb{Z}/101\mathbb{Z}$ such that $\text{ord}_{101}(x) = 10$.

Solution. The modulus $p = 101$ is a prime number, so $\phi(101) = p - 1 = 100$. Let $g = 2$ be the primitive root. Any element $x \in \mathbb{F}_{101}^*$ can be written as $x = g^i = 2^i$ for some exponent $i \in \{1, \dots, 100\}$ [Def 12.2: Primitive Root].

Step 1: Use the Order of a Power Formula The order of a power g^i is given by the formula [Prop 13.1: The Order of a Power]:

$$\text{ord}_p(g^i) = \frac{p-1}{\gcd(p-1, i)}$$

We want $\text{ord}_{101}(x) = 10$, so we set the formula equal to 10:

$$10 = \frac{100}{\gcd(100, i)}$$

Solving for the gcd:

$$\gcd(100, i) = \frac{100}{10} = 10 \quad [\text{Condition on the exponent } i]$$

Step 2: Find a suitable exponent i We need an integer i such that $1 \leq i \leq 100$ and $\gcd(100, i) = 10$. Since $\gcd(100, i) = 10$, i must be a multiple of 10. Let $i = 10k$ for some integer k . Substitute this back into the gcd condition:

$$\gcd(100, 10k) = 10$$

$$10 \cdot \gcd(10, k) = 10 \quad [\text{Property of GCD: } \gcd(na, nb) = n \gcd(a, b)]$$

$$\gcd(10, k) = 1 \quad [\text{Condition on } k]$$

We need to choose a value for k that is coprime to 10. The simplest choice is $k = 1$.

$$k = 1 \implies i = 10 \cdot 1 = 10$$

Step 3: Find the element x We use the exponent $i = 10$ to find the element $x = 2^i$:

$$x = 2^{10} \pmod{101}$$

$$2^{10} = 1024 \quad [\text{Computed value}]$$

$$1024 = 10 \cdot 101 + 14$$

$$x \equiv 14 \pmod{101} \quad [\text{Solution } x]$$

Result: One element with order 10 is $x = 14$. (Other solutions exist for $k = 3, 7, 9$).

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Exercise 6.5

Let $p = 663797$. Which among 3, 7, 10 are primitive roots of $\mathbb{F}_p$?

Solution. The modulus $p = 663797$ is prime. For g to be a primitive root, its order $\text{ord}_p(g)$ must be equal to $p - 1 = 663796$ [Thm 12.2: Characterization of the Primitive Roots].

Step 1: Prime Factorization of $p - 1$

$$p - 1 = 663796$$

$$663796 = 2^2 \cdot 7 \cdot 151 \cdot 157 \quad [\text{Prime factors are } 2, 7, 151, 157]$$

We use the effective test [Lemma 12.1]: g is a primitive root if $g^{(p-1)/q_i} \not\equiv 1 \pmod{p}$ for all prime factors q_i . The key exponents are:

$$E_1 = (p - 1)/2 = 331898$$

$$E_2 = (p - 1)/7 = 94828$$

$$E_3 = (p - 1)/151 = 4396$$

$$E_4 = (p - 1)/157 = 4228$$

We check the candidates by computing $g^{E_1} \pmod{p}$, since this is the largest proper divisor and is often sufficient to show g is not a primitive root (if $g^{E_1} \equiv 1 \pmod{p}$).

Step 2: Test Candidate $g = 3$ We check if $3^{E_i} \equiv 1 \pmod{p}$:

- $3^{331898} \equiv 663796 \equiv -1 \pmod{p}$
- $3^{94828} \equiv 109256 \not\equiv 1 \pmod{p}$
- $3^{4396} \equiv 648796 \not\equiv 1 \pmod{p}$

Since none of the test powers are $1 \pmod{p}$, 3 is a primitive root.

Step 3: Test Candidate $g = 7$ We check $7^{E_1} \pmod{p}$:

- $7^{331898} \equiv 1 \pmod{p}$

Since $7^{(p-1)/2} \equiv 1 \pmod{p}$, $\text{ord}_p(7)$ divides $(p-1)/2 = 331898$. Thus, $\text{ord}_p(7) \leq 331898 < p - 1$. Therefore, 7 is **not** a primitive root.

Step 4: Test Candidate $g = 10$ We check $10^{E_1} \pmod{p}$:

- $10^{331898} \equiv 1 \pmod{p}$

Since $10^{(p-1)/2} \equiv 1 \pmod{p}$, 10 is **not** a primitive root.

Result: Only 3 is a primitive root of $\mathbb{F}_{663797}$.